SMILE²: Estimation–Evolution via Spectral Wave Propagation for Snapshot Compressive Imaging

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Abstract

Coded aperture snapshot spectral imaging (CASSI) reconstructs three-dimensional hyperspectral images from a single two-dimensional coded measurement, resulting in a severely ill-posed spatial–spectral inverse problem. While existing Transformer-based deep networks have achieved impressive performance, their core spatial–spectral interactions rely on implicit token mixing or generic transformations, lacking an explicit account of how information propagates across the hyperspectral field. Moreover, global pairwise interaction incurs quadratic computational complexity. To address these limitations, we propose SMILE² (Spectral Modulated Imaging via Learned Estimation–Evolution), a novel framework that explicitly formulates CASSI reconstruction as a two-stage process: spectral-field estimation and structured field evolution. The estimation stage constructs a dirty shift-back field, a normalized residual-adjoint field and shifted-mask sensing information from the forward model to produce a cleaner initial spectral field. The evolution stage then propagates this estimated field into the final hyperspectral reconstruction. At the core of both stages lies the Spectral Wave Propagator (SWP), which adapts a learnable anisotropic three-dimensional damped-wave model to the sampled spatial–spectral field and performs mask-conditioned global interaction through a closed-form Fourier-domain solution with $O(N \log N)$ operator-level complexity. Extensive experiments on both simulated and real datasets demonstrate that SMILE² achieves state-of-the-art reconstruction quality with compact computation. Ablations and cross-band analyses further validate the complementary roles of field estimation, structured evolution, and spectral propagation.

Code is to be released upon publication.

Introduction

Coded aperture snapshot spectral imaging (CASSI) aims to recover a three-dimensional hyperspectral image (HSI) from a single two-dimensional coded measurement. By using a coded aperture and a dispersive element, CASSI enables efficient snapshot spectral acquisition and avoids slow spatial or spectral scanning (Wagadarikar et al. 2008; Gehm et al. 2007). This optical efficiency, however, comes with a severely underdetermined inverse problem: spatial structures and spectral signatures are modulated, shifted, overlapped, and finally integrated into one compressed measurement.

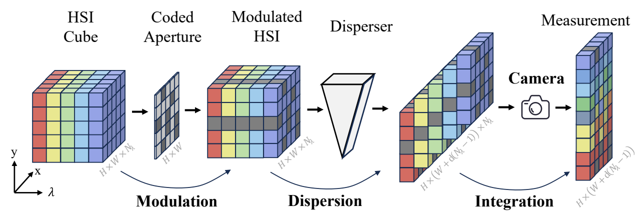

Figure 1: CASSI forward imaging. A spectral cube X is modulated by the coded aperture M, dispersed into shifted spectral measurements, and compressed into a single measurement $Y = \Phi X$.

The CASSI formation process itself suggests that reconstruction should not be treated as a plain image-to-image regression problem. Because coded modulation, wavelength-dependent shift, and spectral integration are entangled in the measurement, a simple shift-back operation can expose coarse scene structures but still leaves a dirty spectral field affected by mask modulation, overlap, uneven sensing coverage, and residual inconsistency. Meanwhile, HSIs are not merely stacks of independent bands: natural spectra often exhibit strong cross-band correlation and low-dimensional material structure (Bioucas-Dias et al. 2012), while spatial edges and textures should remain coherent across wavelengths. These observations point to a field-oriented reconstruction view: estimating a cleaner initial spectral field from the multiplexed measurement, and then propagating information across the sampled spatial–spectral field.

Most end-to-end CASSI methods learn a direct mapping from the compressed measurement to the reconstructed HSI. Early neural models use convolutional or recurrent architectures to aggregate local spatial and spectral information (Miao et al. 2019; Meng, Ma, and Yuan 2020; Cheng et al. 2022), while recent Transformer-based methods further improve reconstruction quality by learning long-range spatial–spectral dependencies (Cai et al. 2022b,a,c; Luo et al. 2025). Attention is powerful because it can learn which spatial regions and spectral bands should communicate. However, this communication is usually expressed as implicit token mixing rather than an explicit propagation rule for disentangling coded spectral overlap and evolving a coherent HSI field. The scaling pressure is also visible in recent end-to-end Transformer models: moving from MST-L to DSMT improves the average peak signal-to-noise ratio (PSNR) from 35.24 dB to 36.92 dB, i.e., a 1.68 dB gain, but increases the parameter count from 2.02M to 12.40M, about 6.1 times larger. Together with the quadratic core complexity of global pairwise interaction, this motivates a more structured and compact alternative for global spatial–spectral modeling.

Recent vision models suggest a complementary route: organizing feature interaction through structured dynamics instead of generic pairwise mixing. vHeat builds vision models upon heat conduction, and WaveFormer introduces wave-equation-based frequency-time visual modeling (Wang et al. 2025b; Shu et al. 2026). Frequency-domain neural operators and token mixers also show that Fourier parameterizations can support efficient global modeling (Li et al. 2021; Guibas et al. 2022; Rao et al. 2021; Lee-Thorp et al. 2022). For CASSI, a purely diffusion-like bias may oversmooth fine spatial textures and spectral variations, whereas a damped wave equation contains propagation, oscillation, and damping. In the Fourier domain, this dynamics admits a closed-form mode-wise evolution rule, making it a natural physics-inspired candidate for evolving a sampled spectral field, without claiming that the reconstructed HSI literally follows the optical propagation inside the CASSI hardware. This leads to the central question:

Can CASSI reconstruction be formulated through a unified spatial–spectral physical operator?

This question is also aligned with a broader movement toward understanding and designing deep models through mathematical structures and dynamical systems. More broadly, recent discussions on a scientific theory of deep learning suggest that mathematical structures and dynamical descriptions can provide useful ways to understand deep models (Simon et al. 2026), while physics-guided machine learning demonstrates the value of embedding scientific structure into neural operators (Karniadakis et al. 2021; Willard et al. 2023; E 2017). This work explores such a direction for CASSI reconstruction.

With that in mind, we present SMILE^2, a single-pass estimation–evolution framework built around a learnable anisotropic wave equation. At its core, SWP adapts a three-dimensional damped-wave model to the sampled HSI field and performs structured global spatial–spectral interaction through a Fourier-domain closed-form solution. CASSI preparation then derives a dirty shift-back field, a measurement-coverage normalization map, a residual back-projection, and shifted-mask information from the forward imaging model. Based on these quantities, SFESTIMATOR (Spectral Field Estimator) estimates a cleaner initial spectral field, and SFEVOLVER (Spectral Field Evolver) further propagates it into the final HSI reconstruction. Compared with the $O(N^2)$ pairwise token mixing of self-attention, SWP provides global interaction through an $O(N \log N)$ operator-level Fourier-domain computation.

Our contributions are summarized as follows.

- We formulate end-to-end CASSI reconstruction through physics-inspired evolution of a discrete spectral field, providing a single-pass alternative to implicit token mixing.
- We design SWP, a mask-conditioned damped-wave propagation operator that performs structured global spatial–spectral interaction through a Fourier-domain closed-form wave kernel.
- We connect the CASSI forward model with spectral-field propagation through CASSI preparation, SFESTIMATOR, and SFEVOLVER, enabling dirty-field correction before structured evolution.
- We build compact SMILE^2-S/M/L variants and obtain state-of-the-art end-to-end reconstruction quality on the TSA simulation benchmark, with favorable spectral angle mapper (SAM) results and mechanism analyses.

Related Work

End-to-End CASSI Reconstruction

Early CASSI reconstruction relies on model-based optimization with hand-crafted priors, which provides an interpretable inverse formulation but requires iterative inference and careful parameter tuning (Bioucas-Dias and Figueiredo 2007; Yuan 2016; Liu et al. 2019). Learning-based reconstruction subsequently develops along direct end-to-end and model-driven unfolding lines. End-to-end methods learn a single-pass mapping from the coded measurement to the HSI, enabling substantially faster inference. Representative designs include convolutional regression in λ-Net, spatial–spectral attention in TSA-Net, bidirectional recurrence in BIRNAT, and high-resolution dual-domain learning in HD-Net (Miao et al. 2019; Meng, Ma, and Yuan 2020; Cheng et al. 2022; Hu et al. 2022).

Transformer-based end-to-end methods further improve reconstruction by modeling long-range spectral correlations. MST performs mask-guided spectral-wise attention, whereas CST and MST++ improve spectral interaction through sparse and multi-stage designs (Cai et al. 2022b,a,c). SPECAT (Yao et al. 2024) progressively accumulates spatial and spectral attention, while S2-Transformer (Wang et al. 2025a) strengthens mask-aware spatial–spectral interaction. DSMT (Luo et al. 2025) adopts coarse-to-fine reconstruction with dual-window multiscale attention, and LSST (Li, Zhao, and Xu 2026) separates lightweight spectral and spatial processing. Despite these advances, spatial–spectral communication is still implemented through learned convolutional or attention-based mixing, without an explicit rule describing how information propagates over the sampled HSI field. SMILE^2 addresses this gap by coupling the x, y, and λ axes through mask-gated damped-wave evolution in the Fourier domain.

Forward-Model Information in End-to-End Reconstruction

Although end-to-end networks learn a direct mapping from measurement to HSI, the CASSI forward model remains essential for network design. GAP-Net embeds projection and back-projection into the network, and DGSMP constructs deep priors from the sensing matrix structure (Meng, Jalali, and Yuan 2020; Huang et al. 2021). These works show that the sensing operator, shifted mask, and measurement-adjoint information provide useful initialization and correction information. Multi-stage unfolding methods further exploit the forward model through repeated data-consistency and learned-prior updates, often achieving higher absolute PSNR at higher iterative cost (Cai et al. 2022d; Zhang et al. 2024a,b). SMILE^2 builds on the broader insight that forward-model information matters in a single-pass end-to-end framework. CASSI preparation condenses measurement-aware quantities into H_0, P, and Q_0, enabling SFESTIMATOR and SFEVOLVER to perform field correction and propagation within one feed-forward reconstruction process.

Frequency- and Physics-Inspired Vision Modeling

Frequency-domain modeling offers a different route to global interaction. Fourier Neural Operator learns solution operators in function spaces and demonstrates that Fourier modes can efficiently parameterize global mappings (Li et al. 2021). FNet mixes tokens with Fourier transforms, GFNet learns global filters, and the Adaptive Fourier Neural Operator (AFNO) adapts neural-operator ideas to high-resolution vision features (Lee-Thorp et al. 2022; Rao et al. 2021; Guibas et al. 2022). In image restoration, FFTformer and frequency-modulated restoration networks use Fourier-domain attention or frequency modulation to model degradation-related components (Kong et al. 2023; Cui et al. 2024). These studies indicate that frequency transforms can provide global receptive fields without explicitly constructing quadratic token affinities.

Another related direction is partial differential equation (PDE)-inspired vision modeling. vHeat builds vision models upon heat conduction, while WaveFormer uses a wave equation to design frequency-time decoupled visual modeling (Wang et al. 2025b; Shu et al. 2026). Compared with total variation, which mainly constrains local first-order gradients, and compared with heat diffusion, which is biased toward smoothing, a damped wave equation contains propagation, oscillation, and damping. This makes it suitable for modeling second-order spatial–spectral structures and frequency-selective attenuation. In spectral reconstruction, Phy-CoSF introduces physics-guided continuous spectral fields for snapshot compressive imaging reconstruction and super-resolution (Chen et al. 2026). However, these frequency- and PDE-inspired works do not directly address CASSI-specific mask modulation, wavelength-dependent shift, 2D-to-3D compression, and shift-back initialization. SMILE^2 differs by coupling CASSI preparation with a mask-gated damped-wave spectral-field propagator in a fixed-grid, single-pass CASSI reconstruction framework.

Method

CASSI Forward Model and Field Preparation

Let $X = [X_1, \dots, X_C] \in \mathbb{R}^{H\times W\times C}$ denote the target HSI sampled at C spectral bands. The coded aperture is represented as $M = [M_1, \dots, M_C] \in \mathbb{R}^{H\times W\times C}$, obtained by broadcasting the physical aperture along the spectral dimension. For the c-th band, $\mathcal{S}_c$ shifts a modulated image horizontally by $d(c-1)$ pixels into the measurement plane. The resulting detector width is $W' = W + d(C-1)$, and the two-dimensional CASSI measurement $Y \in \mathbb{R}^{H\times W'}$ is

$$Y = \Phi X = \sum_{c=1}^{C} \mathcal{S}_c(M_c \odot X_c), \tag{1}$$

where $\Phi : \mathbb{R}^{H\times W\times C} \to \mathbb{R}^{H\times W'}$ denotes the complete mask–shift–sum sensing operator. The contribution support of every band in the detector plane is collected in the shifted-mask tensor

$$M_s = [\mathcal{S}_1(M_1), \dots, \mathcal{S}_C(M_C)] \in \mathbb{R}^{H\times W'\times C}, \tag{2}$$

$$P = \mathrm{diag}(\Phi\Phi^\top) = \sum_{c=1}^{C} M_{s,c}^2 \in \mathbb{R}^{H\times W'}. \tag{3}$$

Thus P is a detector-wise coverage map—the diagonal of the measurement-domain Gram operator—rather than a scalar or a materialized dense matrix.

Let $\mathcal{B}_c$ crop the c-th shifted support from the detector plane and shift it back to the HSI grid. Following the scalar measurement normalization used during training, $\bar{Y} = 2Y/C$, the dirty shift-back field is

$$H_0 = \left[\mathcal{B}_1(\bar{Y}), \dots, \mathcal{B}_C(\bar{Y})\right] \in \mathbb{R}^{H\times W\times C}. \tag{4}$$

Although H_0 recovers the spatial support of each band, it still contains coded-mask modulation, spectral overlap, and dispersion-induced ambiguity. We therefore return the unexplained detector residual to the HSI grid through the normalized adjoint operation

$$R_0 = Y - \Phi H_0, \qquad Q_0 = \Phi^\top \left(\max(P, \epsilon_P)^{-1} \odot R_0\right), \tag{5}$$

where $R_0 \in \mathbb{R}^{H\times W'}$, $Q_0 \in \mathbb{R}^{H\times W\times C}$, and the element-wise maximum with $\epsilon_P = 10^{-6}$ stabilizes division in uncovered detector locations. Neither Φ nor $\Phi^\top$ is materialized; both are evaluated by mask multiplication, shift/crop, back-shift, and spectral summation. The input field to SFESTIMATOR is finally assembled as

$$C_0 = [H_0, Q_0, M, H_0 \odot M] \in \mathbb{R}^{H\times W\times 4C}. \tag{6}$$

This construction keeps field estimation tied to the sensing process: H_0 carries the aliased spectral estimate, Q_0 returns its normalized measurement residual, and M together with $H_0 \odot M$ exposes coded modulation.

Spectral Wave Propagator

SWP is the core physical evolution operator used in both SFESTIMATOR and SFEVOLVER. At its r-th layer, the input $Z_r \in \mathbb{R}^{H_r\times W_r\times C_r}$ is interpreted as a latent spatial–spectral field over (x, y, λ), where λ denotes the spectral coordinate and is not used as a discrete band index. The propagation variable t denotes feature-field evolution depth rather than physical exposure time in the CASSI optical path. Thus, SWP uses wave dynamics as a physics-inspired structure

Figure 2: Conceptual overview of the transition from attention mixing to physical evolution. SMILE[2] replaces generic pairwise token interaction with structured spatial–spectral propagation governed by a Fourier-domain damped-wave kernel.

for feature propagation, while its propagation parameters remain data-driven.

A local depth-wise projection followed by a point-wise channel projection splits the normalized input into a wave branch F_r^w and a content branch F_r^g:

$$(F_r^w, F_r^g) = \mathrm{Split}\left[\Pi_r^{\mathrm{in}}(\mathrm{DWConv}_{3\times 3}(\mathrm{LN}(Z_r)))\right]. \tag{7}$$

Both branches belong to $\mathbb{R}^{H_r\times W_r\times C_r}$. The alignment operator $\mathcal{A}_r$ first crops the detector-width tensor M_s to the reconstruction support and then uses sigmoid-activated, learned strided projections to match the spatial and channel resolution of layer r. Its soft amplitude gate is

$$g_r = \epsilon_M + (1-\epsilon_M)\mathcal{A}_r(M_s), \qquad g_r \in \mathbb{R}^{H_r\times W_r\times C_r}, \tag{8}$$

where $\epsilon_M = 0.1$ retains a nonzero propagation amplitude at fully masked locations. Two depth-wise/point-wise projections then generate the initial displacement and velocity:

$$u_0 = \Pi_u(F_r^w) \odot g_r, \qquad v_0 = \Pi_v(F_r^w) \odot g_r, \tag{9}$$

where $u_0, v_0 \in \mathbb{R}^{H_r\times W_r\times C_r}$. A 3D Fourier transform over the two spatial axes and the spectral axis gives

$$\hat{u}_0 = \mathcal{F}_{x,y,\lambda}(u_0), \qquad \hat{v}_0 = \mathcal{F}_{x,y,\lambda}(v_0). \tag{10}$$

The Fourier transform along λ is used as a discrete spectral-mode basis over the sampled HSI bands; it does not assume that real spectral reflectance is physically periodic. In implementation, the spectral axis is zero-padded to the next power of two before the real-valued three-dimensional fast Fourier transform (3D FFT) and cropped back after inverse transformation.

Let $u(x, y, \lambda, t)$ denote the continuous field whose samples are carried by the wave branch. The continuous form underlying SWP is an anisotropic damped wave equation on this spatial–spectral field:

$$\partial_{tt}u + \alpha\partial_t u = v_s^2(\partial_{xx} + \partial_{yy})u + v_\lambda^2\partial_{\lambda\lambda}u. \tag{11}$$

Here v_s controls spatial propagation, v_λ controls spectral propagation, and α controls damping. To model frequency-selective propagation, SWP generalizes the prototype scalar α and v_s to spectral-mode-dependent functions $\alpha(f_\lambda)$ and $v_s(f_\lambda)$. For a Fourier mode $\boldsymbol{f} = (f_x, f_y, f_\lambda)$, the spatial and spectral derivatives become multiplicative frequency responses. The PDE is therefore reduced to an independent second-order ordinary differential equation for each mode:

$$\frac{d^2\hat{u}(\boldsymbol{f},t)}{dt^2} + \alpha(f_\lambda)\frac{d\hat{u}(\boldsymbol{f},t)}{dt} + \omega_0^2(\boldsymbol{f})\hat{u}(\boldsymbol{f},t) = 0, \tag{12}$$

with initial conditions $\hat{u}(\boldsymbol{f},0) = \hat{u}_0(\boldsymbol{f})$ and $\partial_t\hat{u}(\boldsymbol{f},0) = \hat{v}_0(\boldsymbol{f})$. The undamped angular frequency and discriminant are

$$\begin{aligned} \omega_0^2(\boldsymbol{f}) &= (2\pi)^2\left[v_s^2(f_\lambda)(f_x^2+f_y^2) + v_\lambda^2 f_\lambda^2\right], \\ \eta(\boldsymbol{f}) &= \omega_0^2(\boldsymbol{f}) - \left(\frac{\alpha(f_\lambda)}{2}\right)^2. \end{aligned} \tag{13}$$

The per-mode closed-form solution is

$$\begin{aligned} \hat{u}(\boldsymbol{f},t) =& e^{-\alpha(f_\lambda)t/2}\Big[\hat{u}_0(\boldsymbol{f})\mathrm{Cs}(\eta(\boldsymbol{f}),t) \\ &+ \left(\hat{v}_0(\boldsymbol{f}) + \frac{\alpha(f_\lambda)}{2}\hat{u}_0(\boldsymbol{f})\right)\mathrm{Sn}(\eta(\boldsymbol{f}),t)\Big]. \end{aligned} \tag{14}$$

The functions Cs and Sn compactly cover the under-damped, critically damped, and over-damped cases:

$$\mathrm{Cs}(\eta,t) = \begin{cases} \cos(\sqrt{\eta}t), & \eta > 0, \\ 1, & \eta = 0, \\ \cosh(\sqrt{-\eta}t), & \eta < 0, \end{cases} \tag{15a}$$

$$\mathrm{Sn}(\eta,t) = \begin{cases} \dfrac{\sin(\sqrt{\eta}t)}{\sqrt{\eta}}, & \eta > 0, \\ t, & \eta = 0, \\ \dfrac{\sinh(\sqrt{-\eta}t)}{\sqrt{-\eta}}, & \eta < 0. \end{cases} \tag{15b}$$

The evolved spatial–spectral field is recovered by inverse FFT:

$$u(t) = \mathcal{F}_{x,y,\lambda}^{-1}(\hat{u}(\boldsymbol{f},t)). \tag{16}$$

In implementation, $\alpha(f_\lambda)$ and $v_s(f_\lambda)$ are layer-specific learnable vectors indexed by spectral modes, while v_λ and t are learnable scalars; the layer index is omitted above for readability. All four quantities are passed through a positivity mapping before use. This gives SWP spectral-mode-dependent damping and propagation speed without claiming wavelength-specific physical coefficients in the optical hardware. After inverse FFT, the evolved field is normalized and modulated by $\Gamma_r(F_r^g)$, where Γ_r is a point-wise projection followed by an 11×11 depth-wise convolution that supplies spatial-content modulation. The result is then projected back to the feature dimension. The complete residual update is

$$\mathcal{W}_{\theta_r}(Z_r, M_s) = Z_r + \Pi_o(\mathrm{LN}(u(t)) \odot \Gamma_r(F_r^g)), \quad (17)$$

where Π_o is the output projection and θ_r collects the learnable parameters of layer r. For $N = H_r W_r C_r$ feature samples, FFT and point-wise per-mode evolution require $O(N \log N)$ operations, whereas explicitly forming all pairwise token relations requires $O(N^2)$.

Spectral Field Estimator

The aliased shift-back field H_0 is not directly suitable as the final reconstruction. SFESTIMATOR first predicts a zero-initialized residual correction:

$$\Delta_\psi(C_0) = \mathrm{Conv}^0_{3\times3}(\sigma(\mathrm{Conv}_{3\times3}(C_0))), \quad (18)$$

where σ is GELU and $\mathrm{Conv}^0_{3\times3}$ denotes a zero-initialized 3×3 convolution. The correction $\Delta_\psi(C_0)$ has the same $H \times W \times C$ dimensions as H_0. The resulting estimator input is

$$Z_0 = H_0 + \Delta_\psi(C_0), \qquad Z_0|_{\mathrm{init}} = H_0, \quad (19)$$

because the last convolution of Δ_ψ is initialized to zero. The first SWP-based U-Net, parameterized by θ_E, estimates a cleaner spectral field:

$$U_1 = \mathcal{E}^{\mathrm{SWP}}_{\theta_E}(Z_0, M_s). \quad (20)$$

Spectral Field Evolver

SFEVOLVER does not recompute R_0 or Q_0. Instead, it further evolves the estimated spectral field U_1:

$$U_2 = \mathcal{E}^{\mathrm{SWP}}_{\theta_V}(U_1, M_s), \qquad \hat{X} = U_2. \quad (21)$$

Here θ_V denotes the evolver parameters, and $U_1, U_2 \in \mathbb{R}^{H\times W\times C}$.

Experiments

Experimental Settings

Simulation experiments are conducted on the TSA simulation benchmark with ten test scenes of spatial size 256×256 and $C = 28$ spectral channels. Training data are sampled from CAVE and KAIST hyperspectral datasets (Yasuma et al. 2010; Choi et al. 2017). We report PSNR, structural similarity (SSIM), and SAM, where SAM measures spectral directional error and is important for hyperspectral fidelity (Kruse et al. 1993). All training and evaluation experiments are conducted on NVIDIA A800 GPUs with 80GB memory. The main models are trained for 300 epochs with Adam and cosine learning-rate decay. We instantiate three compact variants under different computational budgets. SMILE^2-S uses the $[2, 2, 2]$ backbone with partially tied estimator/evolver parameters, SMILE^2-M decouples their parameters at the same width, and SMILE^2-L increases the U-Net width to $[2, 4, 4]$. We report model complexity using parameter count and floating-point operations (FLOPs).

SFESTIMATOR and SFEVOLVER output U_1 and $U_2 = \hat{X}$, respectively. Training uses only weighted root mean square error (RMSE) reconstruction supervision:

$$\mathcal{L}_{\mathrm{train}} = \sum_{k=1}^{2} \beta_k \,\mathrm{RMSE}(U_k, X), \qquad \boldsymbol{\beta} = [0.2, 1.0]. \quad (22)$$

The objective does not include auxiliary SAM/SSIM, spectral-derivative, spectral-angle, or explicit $\|\Phi\hat{X} - Y\|$ data-fidelity losses. Forward-model consistency enters before reconstruction through CASSI preparation: H_0 supplies the dirty shift-back field, Q_0 exposes the normalized residual-adjoint field, and P represents the measurement-domain coding coverage.

Quantitative Comparison

Table 1 reports per-scene results on the TSA simulation benchmark. For methods with released weights or reproducible official implementations, we evaluate them under the same protocol; for methods without released weights or reconstructed outputs, PSNR/SSIM are taken from the corresponding papers and SAM is denoted by "–". Compared with earlier convolutional and Transformer end-to-end baselines, SMILE^2 consistently improves reconstruction quality while keeping the model compact. SMILE^2-M uses 2.00M parameters and 28.28G FLOPs, reaching 36.57 dB PSNR, 0.961 SSIM, and 0.097 rad SAM. Scaling the same estimation–evolution principle to SMILE^2-L improves the average score to 37.07 dB PSNR, 0.965 SSIM, and 0.086 rad SAM with 3.19M parameters and 38.60G FLOPs. These results indicate that structured spectral-field evolution can serve as an effective end-to-end reconstruction prior, improving both image-level fidelity and spectral-angle consistency.

Qualitative Results

Figures 5 and 6 show representative simulation results. In Scene 5, low-accuracy methods recover the rough layout of the "SIG" region but introduce conspicuous artifacts and spectral distortion. SMILE^2-L preserves a more continuous text contour and a stable structure across the displayed bands. DSMT retains slightly finer local texture in parts of the enlarged region, whereas SMILE^2-L emphasizes cleaner boundaries and a spectral density curve that follows the ground-truth trend closely. Scene 7 contains stronger spatial–spectral contrast. SMILE^2-L maintains coherent petal silhouettes across wavelengths and suppresses dispersion-like artifacts, while DSMT again preserves finer high-frequency detail in some local regions. These comparisons indicate that the main qualitative advantage of SMILE^2-L lies in contour continuity and inter-band structural consistency rather than uniformly sharper texture.

We also apply SMILE^2 to real CASSI measurements. As shown in Fig. 7, SMILE^2 produces stable reconstructions on real coded measurements. Compared with other end-to-end baselines, it shows fewer artifacts in smooth regions and

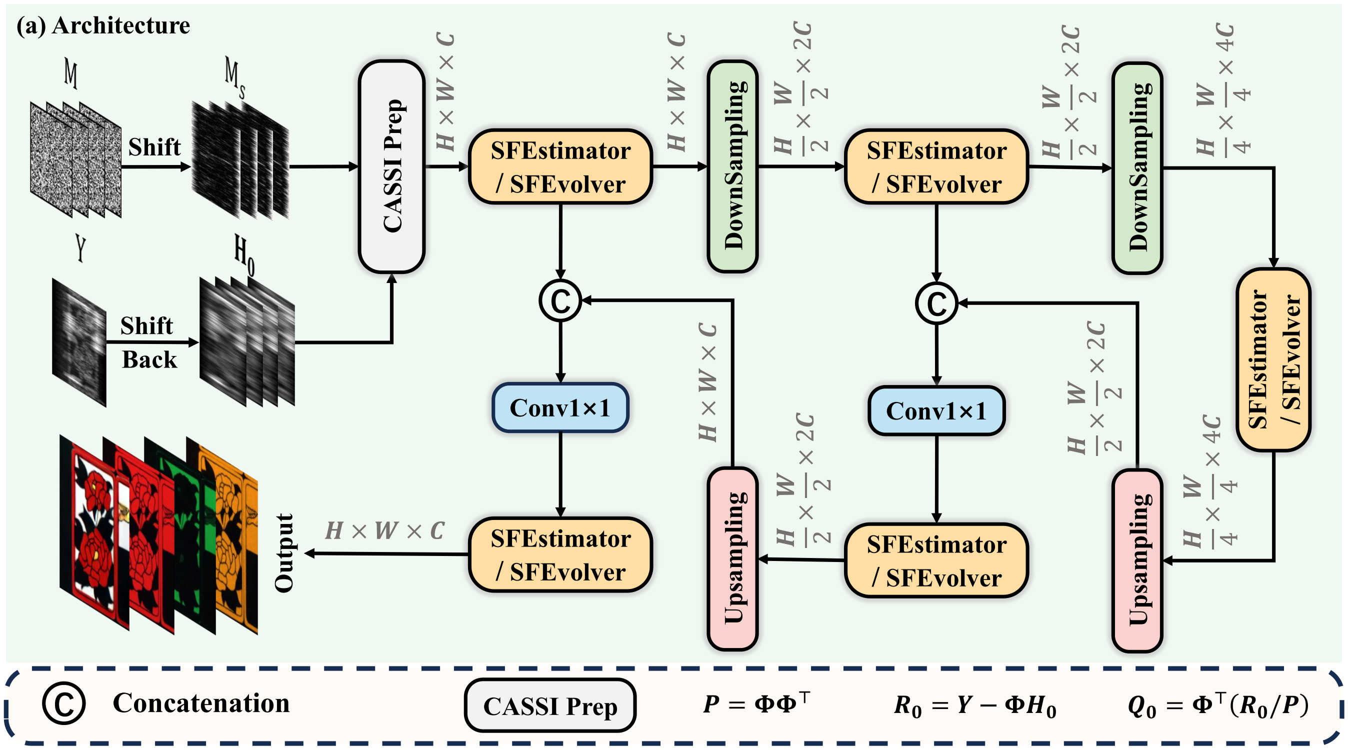

Figure 3: Overall architecture of SMILE². CASSI preparation constructs H_0, Q_0, and the shifted-mask tensor M_s from the measurement and mask. SFESTIMATOR estimates the initial spectral field U_1, and SFEVOLVER evolves it into the final reconstruction $U_2 = \hat{X}$.

clearer local edges, suggesting that the estimation–evolution structure generalizes to real measurements.

Mechanism Analysis

We analyze the explicitness of SMILE² from three complementary levels.

Operator-form level. SWP does not represent global interaction as an unconstrained affinity matrix. Its mixing rule is written as a mask-conditioned damped-wave evolution in the Fourier domain, where damping, spatial propagation speed, spectral propagation speed, and evolution time enter the closed-form kernel with specific roles. This gives the core interaction an explicit operator form, although the surrounding feature projections and nonlinear mappings remain learned neural modules.

Propagation-direction level. We examine whether the trained model uses structured cross-band information. We estimate a finite-difference cross-band response map and compare equal-count occlusion of neighboring and distant bands. As shown in Fig. 8, the response is dominated by the diagonal and nearby wavelengths, while still preserving weak long-range spectral interactions. This pattern is consistent with the spectral continuity and material-dependent variation of HSIs: neighboring bands are usually more redundant, but distant bands can still provide complementary context.

Intervention level. We further perturb the learned SWP dynamics by changing the interpretable propagation parameters, with visual results shown in Fig. 9. Starting from the learned response, increasing damping reduces the response energy from 13.04 to 10.06 and the center-band peak from 0.123 to 0.064, while lowering damping increases them to 16.32 and 0.171. Increasing spatial speed expands the response and raises the total energy to 18.27; increasing spectral speed changes the wavelength-axis response and lowers the center-band peak to 0.096. A longer evolution time also broadens the response profile, increasing the spectral spread from 6.88 to 7.28. These controlled changes support the intervention-level claim: SMILE² is not fully white-box, but its central propagation rule is explicit in form, visualizable in direction, and causally controllable through physically inspired parameters.

Ablation Study

Table 2 quantifies the contribution of each component. Removing SFESTIMATOR lowers PSNR by 0.44 dB and SSIM by 0.0075, while increasing SAM from 0.1116 to 0.1352; the measurement-aware initial-field correction is therefore beneficial before evolution. Removing SFEVOLVER causes the largest degradation, with an 11.60 dB PSNR drop and a SAM increase to 0.4790, confirming that the estimated field alone is not the final reconstruction and requires structured evolution. Without shifted-mask gating, PSNR decreases by

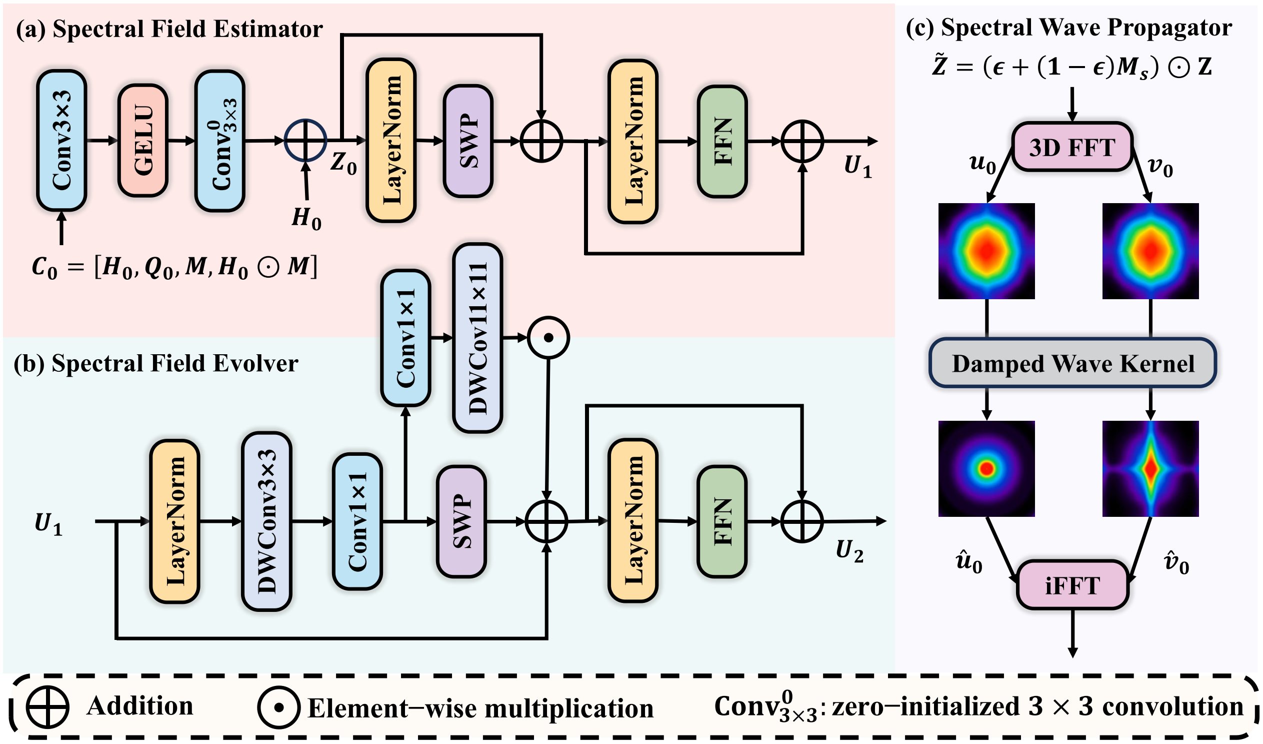

Figure 4: Signal flow of SFEstimator, SFEvolver, and the embedded SWP operator. The SWP branch performs mask-gated Fourier-domain damped-wave propagation, while SFEstimator and SFEvolver differ in their input fields and update roles.

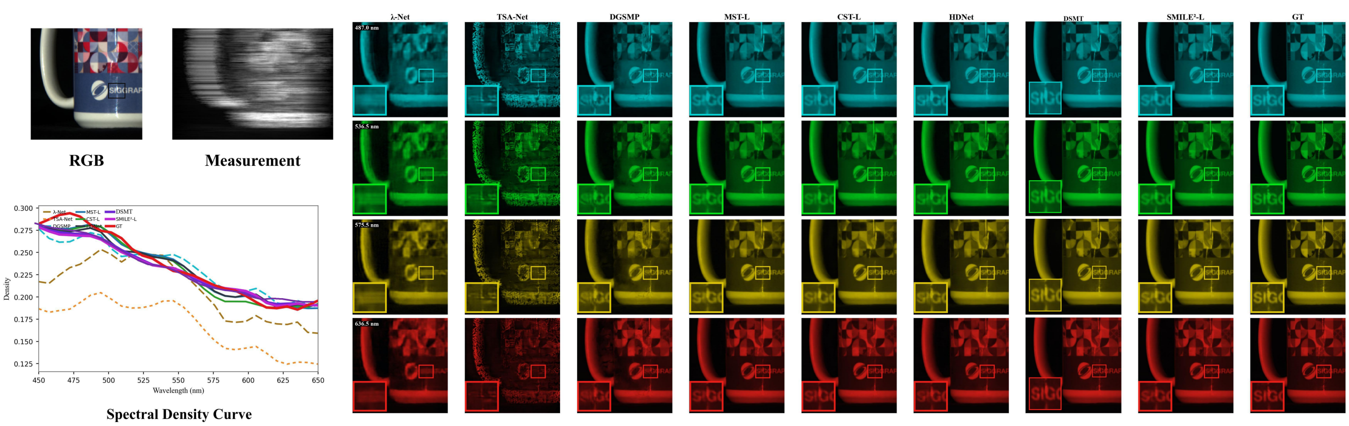

Figure 5: Simulation Scene 5. The figure compares red-green-blue (RGB) reference and measurement inputs, representative spectral bands, zoomed local regions, and spectral density curves.

0.57 dB and SAM rises to 0.1517, the largest spectral-angle penalty among the non-degenerate variants. Suppressing the λ-derivative similarly reduces PSNR by 0.44 dB and increases SAM to 0.1345, supporting the role of explicit spectral-axis propagation.

Discussion

What is explicit in SMILE^2. Compared with Transformer-style interaction, SMILE^2 makes spatial–spectral communication more explicit at three levels. The operator form is specified by the Fourier-domain damped-wave kernel; the propagation direction can be inspected through impulse and cross-band responses; and the behavior can be intervened on by changing damping, spatial speed, spectral speed, or evolution time. This is the sense in which SMILE^2 is more explicit than implicit token mixing. It should not be interpreted as a completely transparent white-box model, because the input and output projections, nonlinear feature mappings, and learned parameter generation are still data-driven.

Table 1: Quantitative comparison on the TSA simulation benchmark. Each scene cell reports PSNR, SSIM, and SAM (rad) from top to bottom. Methods with public checkpoints or official reproducible implementations are evaluated under the same protocol; for methods without released weights or reconstructed outputs, PSNR/SSIM are taken from the corresponding papers and SAM is unavailable. CNN and MAP denote convolutional neural network and maximum a posteriori, respectively.

Algorithm	Method	Params	GFLOPs	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Avg
Optimization	TwIST (Bioucas-Dias and Figueiredo 2007)	–	–	25.16 0.700 –	23.02 0.604 –	21.40 0.711 –	30.19 0.851 –	21.41 0.635 –	20.95 0.644 –	22.20 0.643 –	21.82 0.650 –	22.42 0.690 –	22.67 0.569 –	23.12 0.669 –
Optimization	GAP-TV (Yuan 2016)	4.27	75.63	21.26 0.490 0.553	21.90 0.578 0.618	18.97 0.506 0.561	27.97 0.794 0.455	17.43 0.429 0.504	21.88 0.688 0.548	20.65 0.442 0.530	22.19 0.700 0.580	22.27 0.609 0.366	19.94 0.615 0.594	21.44 0.585 0.531
Optimization	DeSCI (Liu et al. 2019)	–	–	27.13 0.748 –	23.04 0.620 –	26.62 0.818 –	34.96 0.897 –	23.94 0.706 –	22.38 0.683 –	24.45 0.743 –	22.03 0.673 –	24.56 0.732 –	23.59 0.587 –	25.27 0.721 –
CNN	λ-Net (Miao et al. 2019)	32.73	28.40	29.71 0.832 0.244	27.70 0.746 0.295	29.55 0.852 0.260	37.58 0.919 0.392	26.61 0.804 0.260	27.28 0.809 0.403	26.62 0.783 0.244	26.23 0.807 0.416	28.56 0.805 0.265	26.16 0.723 0.391	28.60 0.808 0.317
CNN	TSA-Net (Meng, Ma, and Yuan 2020)	44.25	73.83	24.42 0.714 0.256	23.38 0.660 0.315	17.30 0.614 0.445	26.34 0.795 0.415	20.51 0.600 0.275	22.29 0.742 0.368	21.35 0.636 0.374	22.25 0.762 0.395	20.27 0.672 0.446	21.29 0.694 0.442	21.94 0.689 0.373
MAP+CNN	DGSMP (Huang et al. 2021)	3.76	646.36	33.35 0.920 0.161	31.66 0.892 0.208	32.92 0.925 0.135	40.39 0.970 0.135	29.46 0.894 0.172	32.74 0.938 0.144	31.14 0.898 0.141	31.33 0.932 0.190	31.53 0.925 0.145	31.51 0.934 0.129	32.60 0.923 0.156
CNN	HDNet (Hu et al. 2022)	2.37	154.73	35.11 0.940 0.129	35.65 0.943 0.142	36.05 0.948 0.099	42.48 0.978 0.101	32.67 0.950 0.089	34.47 0.956 0.120	33.64 0.930 0.114	32.45 0.948 0.141	34.85 0.947 0.110	32.34 0.943 0.119	34.97 0.948 0.116
Transformer	MST-L (Cai et al. 2022b)	2.02	25.67	35.44 0.946 0.123	36.12 0.949 0.142	36.39 0.955 0.106	42.05 0.977 0.128	32.94 0.950 0.101	34.71 0.957 0.137	34.08 0.932 0.110	32.90 0.953 0.179	35.04 0.947 0.128	32.74 0.946 0.146	35.24 0.951 0.130
Transformer	CST-L (Cai et al. 2022a)	3.00	20.34	35.82 0.951 0.116	36.52 0.955 0.118	37.45 0.963 0.082	42.46 0.979 0.100	33.34 0.956 0.078	35.48 0.966 0.104	34.40 0.940 0.098	33.80 0.965 0.119	35.88 0.957 0.100	33.35 0.955 0.103	35.85 0.959 0.102
Transformer	S2Transformer (Wang et al. 2025a)	1.80	27.21	36.17 0.949 –	37.57 0.958 –	37.29 0.957 –	42.96 0.975 –	34.40 0.960 –	36.44 0.965 –	35.41 0.946 –	34.50 0.962 –	36.54 0.959 –	33.57 0.952 –	36.48 0.958 –
Transformer	DSMT (Luo et al. 2025)	12.40	44.94	36.38 0.958 0.103	37.86 0.965 0.104	39.18 0.968 0.064	43.89 0.985 0.090	34.03 0.965 0.071	36.48 0.972 0.094	35.48 0.952 0.088	34.54 0.969 0.108	37.67 0.966 0.083	33.71 0.959 0.090	36.92 0.966 0.090
Wave equation	SMILE^2-S	1.02	28.28	35.68 0.946 0.114	36.39 0.950 0.131	37.39 0.961 0.084	44.07 0.984 0.104	33.42 0.955 0.086	35.20 0.961 0.123	34.29 0.936 0.101	33.27 0.957 0.148	37.21 0.960 0.102	32.45 0.947 0.124	35.94 0.956 0.112
Wave equation	SMILE^2-M	2.00	28.28	36.16 0.952 0.106	37.43 0.959 0.113	38.77 0.967 0.070	43.67 0.982 0.089	33.86 0.960 0.074	35.66 0.965 0.106	35.13 0.947 0.090	33.98 0.964 0.128	37.93 0.965 0.089	33.09 0.953 0.104	36.57 0.961 0.097
Wave equation	SMILE^2-L	3.19	38.60	36.60 0.957 0.091	38.04 0.966 0.097	39.25 0.969 0.071	44.78 0.987 0.080	34.23 0.964 0.072	35.94 0.969 0.089	35.61 0.952 0.078	33.96 0.966 0.103	38.65 0.969 0.082	33.59 0.956 0.097	37.07 0.965 0.086

Physical meaning and boundary. The spectral field in SMILE^2 is a discretely sampled HSI feature field over (x, y, λ), not a continuous material wave measured in the optical path. Accordingly, v_s, v_λ, α, and t should be understood as learned feature-field propagation parameters rather than true optical constants. Within this boundary, the damped-wave formulation provides a structured prior: it couples spatial and spectral second-order variation, permits long-range Fourier-domain propagation, and introduces damping to prevent unconstrained oscillatory mixing. This distinguishes SWP from a free Fourier filter and from attention affinity, while keeping the claim physics-inspired rather than physics-exact.

Inference latency and memory. Table 3 reports inference latency and peak allocated memory under the same test protocol. All methods are tested scene-by-scene at $256 \times 256 \times 28$ resolution. These measurements complement FLOPs and parameter counts, since actual throughput also depends on kernel implementation, memory access, and FFT scheduling.

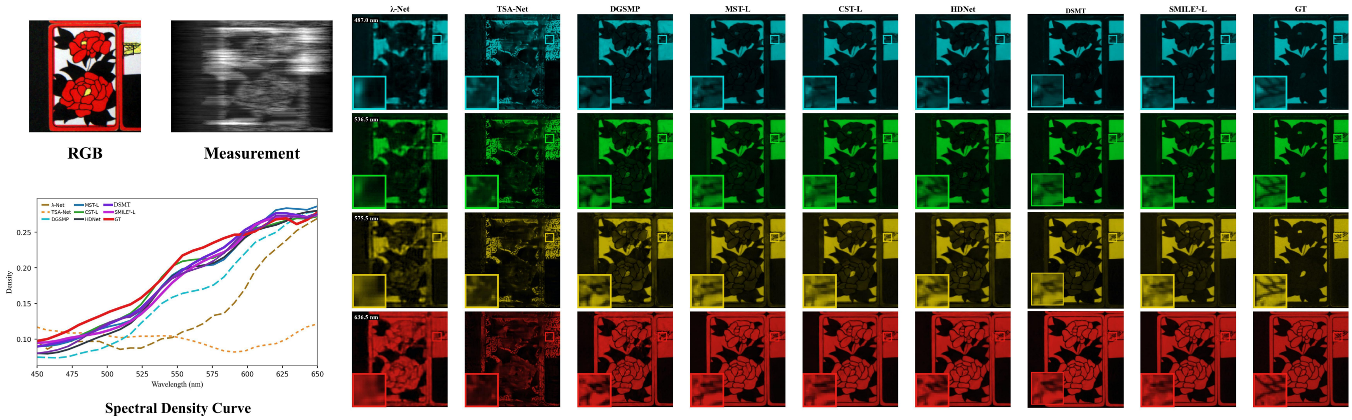

Figure 6: Simulation Scene 7. The same set of end-to-end comparison methods and curve styles are used as in Scene 5.

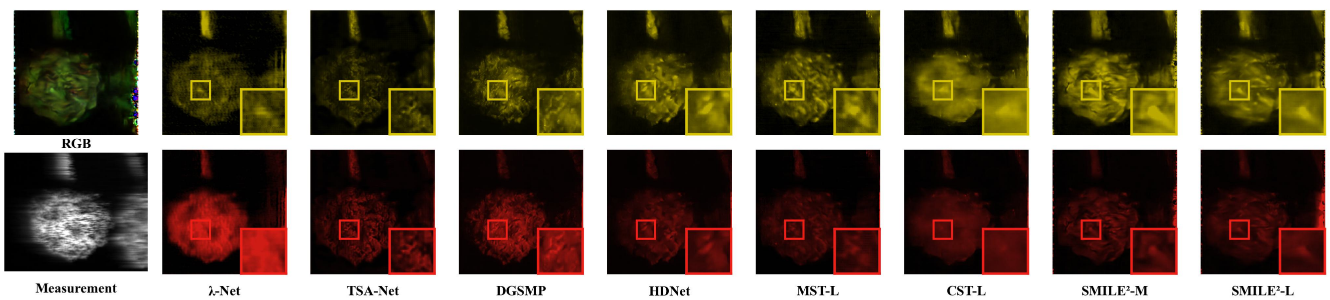

Figure 7: Real CASSI measurement results. Since real measurements do not provide ground-truth HSIs, we compare representative bands and zoomed local regions.

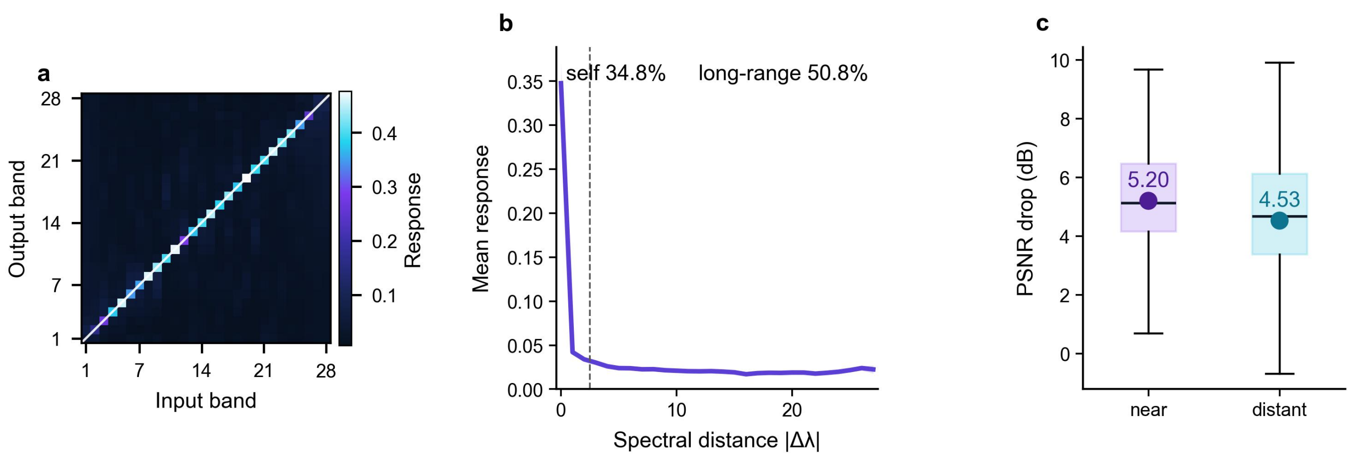

Figure 8: Mechanism analysis of cross-band propagation. The finite-difference mixing map and equal-count occlusion results reveal how SMILE^2 exploits spectral context.

Learned propagation profiles. Figure 11 visualizes the learned SWP parameters in SMILE^2-M. $\alpha(f_\lambda)$ and $v_s(f_\lambda)$ are computed from 28-mode SWP layers and thus correspond to spectral-mode-dependent damping and propagation speed. v_λ and t are module-level learnable scalars that control spectral second-derivative strength and evolution depth. Figure 12 further derives 3D damped-wave kernel visualizations from the learned average parameters under different evolution depths. Together with the controlled interventions in Fig. 9, these profiles show that the learned operator does not collapse to a constant response and that its propagation behavior can be inspected at both parameter and response levels.

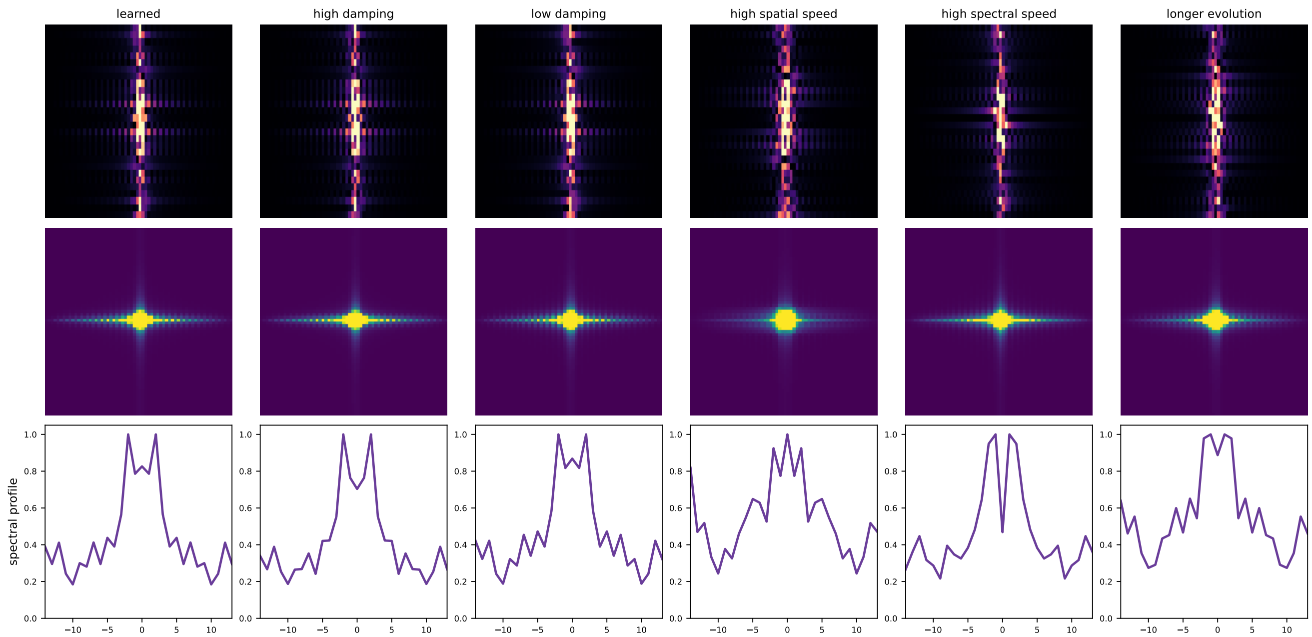

Figure 9: Controlled intervention on the learned SWP operator in SMILE^2-M. A spatial–spectral impulse is propagated by the closed-form damped-wave kernel while perturbing damping, spatial speed, spectral speed, and evolution time. These responses are generated by the explicit operator without retraining.

Table 2: Ablation study under the SMILE^2-S setting.

ID	Variant	PSNR↑	SSIM↑	SAM↓
B0	SMILE^2-S	35.9378	0.9557	0.1116
A1	w/o SFEstimator	35.4985	0.9482	0.1352
A2	w/o SFEvolver	24.3386	0.6160	0.4790
A3	w/o shifted-mask gating	35.3727	0.9473	0.1517
A4	w/o λ-derivative	35.5002	0.9484	0.1345

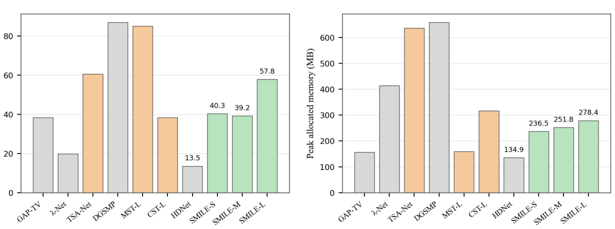

Figure 10: Latency and peak-memory comparison under the same test protocol.

Table 3: Inference latency and peak allocated memory on the TSA simulation benchmark.

Method	Params (M)	GFLOPs (G)	Time (ms)	Alloc. (MB)	Reserved (MB)
GAP-Net	4.27	75.63	38.23	156.17	194.00
λ-Net	32.73	28.40	19.71	414.00	500.00
TSA-Net	44.25	73.83	60.47	635.97	784.00
DGSMP	3.76	646.36	86.89	658.29	774.00
MST-L	2.02	25.67	84.96	158.76	274.00
CST-L	3.00	20.34	38.25	316.45	388.00
HDNet	2.37	154.73	13.49	134.94	178.00
SMILE^2-S	1.02	28.28	40.26	236.47	274.00
SMILE^2-M	2.00	28.28	39.16	251.79	290.00
SMILE^2-L	3.19	38.60	57.82	278.42	316.00

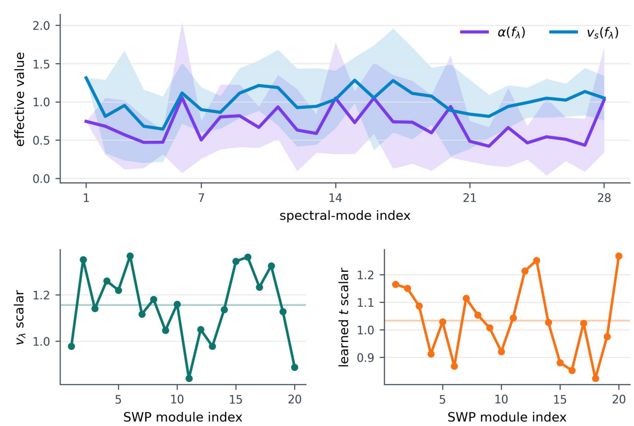

Figure 11: Learned spectral-mode parameters in SMILE^2-M.

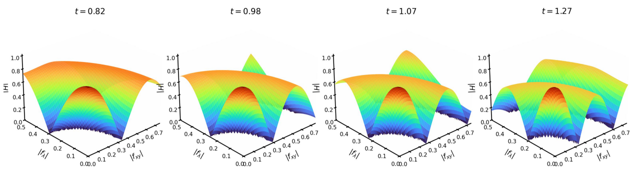

Figure 12: Derived 3D visualization of the learned damped-wave kernel under different evolution depths t.

Conclusion

We presented SMILE^2, a single-pass spectral-field estimation–evolution framework for CASSI reconstruction. SMILE^2 constructs measurement-aware CASSI fields, estimates a cleaner initial spectral field, and evolves it through a Fourier-domain damped-wave propagator. Experiments on simulation and real CASSI measurements demonstrate state-of-the-art end-to-end reconstruction quality with compact computation. Mechanism analyses further show that SMILE^2 exploits structured cross-band propagation, providing a physically motivated alternative to generic token mixing for hyperspectral reconstruction.

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